

Azure Data Factory Real Time Scenario Part - 20

**How to Copy Only Matching Files
From Azure Data Lake to SQL Server**

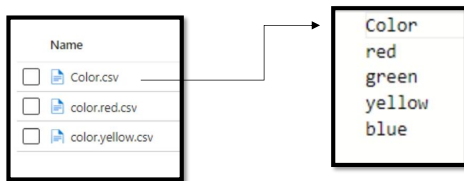
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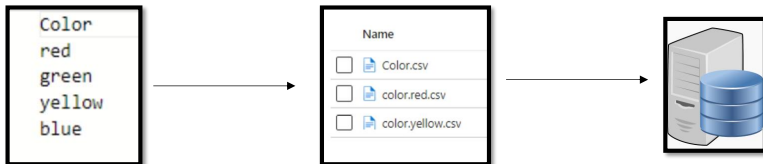
Praveen Patel
Azure Data Engineer

Business Requirement

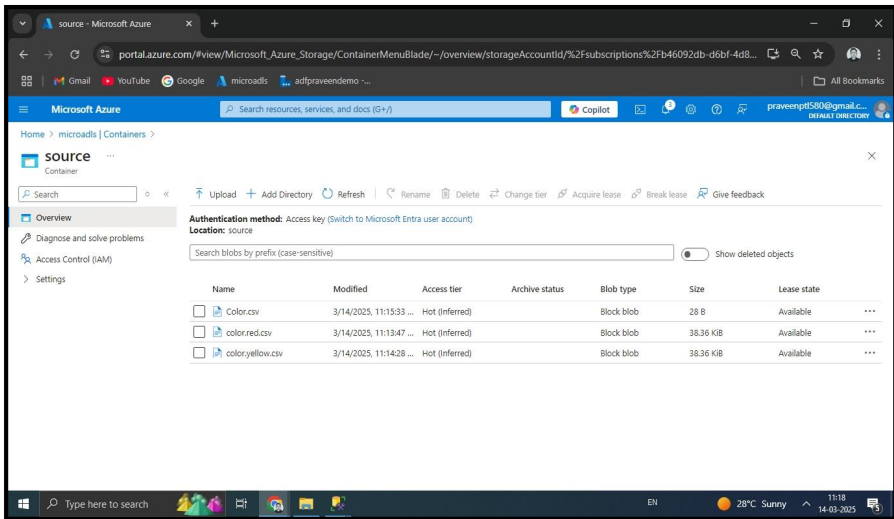
Suppose i have source container in ADLS account that contains Color.csv file, color.red.csv file and color.yellow.csv file. Color.csv file contains a single column 'Color' having values red, green, yellow and blue.



As per business requirement we will check do values contain in Color column match with .csv filename that means Color column values exist in .csv filename substring or not. If it exists we copy those .csv files to the on premise sql server.



As i discussed earlier in the business requirement i have source container in ADLS account having Color.csv file, color.red.csv and color.yellow.csv file.

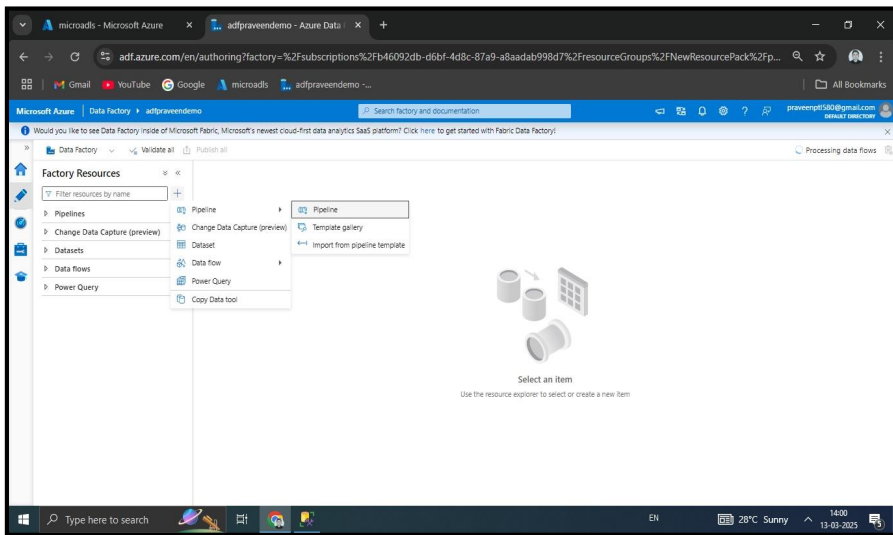


The screenshot displays the Microsoft Azure portal interface for a storage container named 'source'. The left sidebar shows navigation options: Overview, Diagnose and solve problems, Access Control (IAM), and Settings. The main content area shows the container's details, including the authentication method (Access key) and location (source). A table lists the files in the container:

	Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
☐	Color.csv	3/14/2025, 11:15:33 ...	Hot (Inferred)		Block blob	28 B	Available ...
☐	color.red.csv	3/14/2025, 11:13:47 ...	Hot (Inferred)		Block blob	38.36 KiB	Available ...
☐	color.yellow.csv	3/14/2025, 11:14:28 ...	Hot (Inferred)		Block blob	38.36 KiB	Available ...

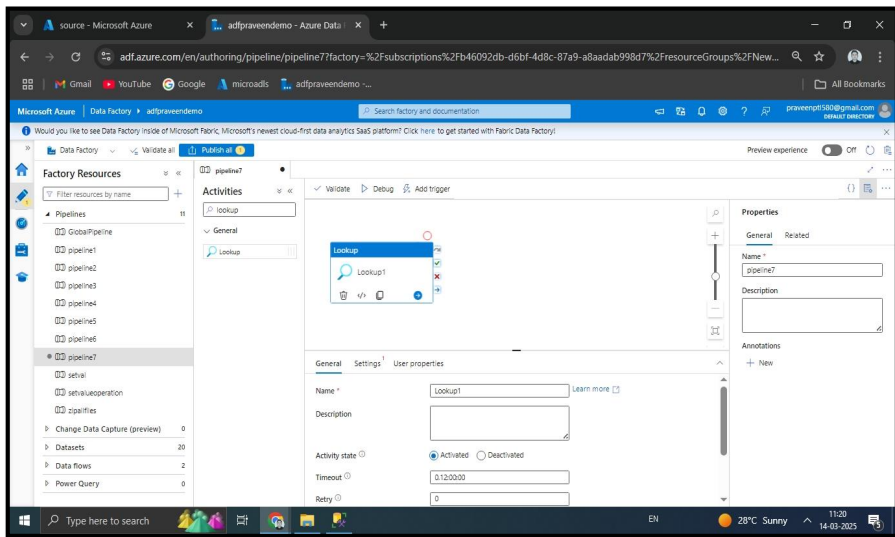
Step 1

Open Azure Factory go to Azure Data Factory Author Tab click on '+' icon and create a new pipeline.



Step 2

Next go to activities search bar search and select look up activity to read the content of Color.csv file then drag and drop it.



Step 3

Go to lookup activity setting click on '+ New' to create a new dataset.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline7' selected. The main canvas shows the 'Lookup' activity configuration for 'pipeline7'. The 'Activities' pane on the left lists 'lookup' and 'Lookup'. The 'Properties' pane on the right shows the 'General' tab with the 'Name' field set to 'pipeline7'. The 'Settings' tab is active, showing the 'Source dataset' dropdown set to 'Select...' and the 'First row only' checkbox checked. The 'User properties' tab is also visible. The bottom status bar shows the time as 11:20 on 14-03-2023.

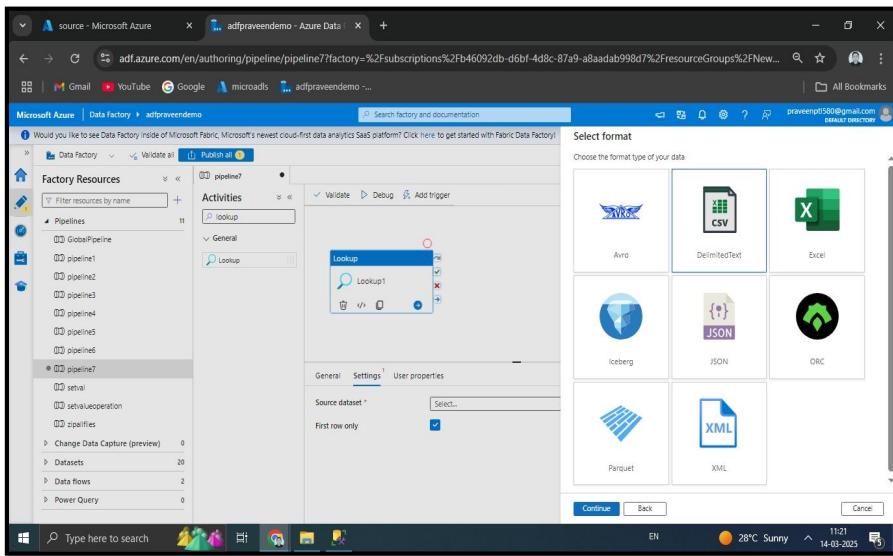
Step 4

Search and Select ADLS Gen2 for new dataset then proceed with continue.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline7' selected. The 'Activities' pane for 'pipeline7' shows a 'Lookup' activity named 'Lookup1'. The 'Settings' tab for 'Lookup1' is active, showing the 'Source dataset' dropdown set to 'Select...' and the 'First row only' checkbox checked. On the right, the 'New dataset' dialog is open, showing the 'Azure Data Lake' search results. The 'All' tab is selected, and 'Azure Data Lake Storage Gen2' is highlighted. The 'Continue' button is visible at the bottom of the dialog. The top of the browser window shows the URL 'adf.azure.com/en/authoring/pipeline/pipeline7?factory=%2Fsubscriptions%2Fb46092db-d6bf-4d8c-87a9-a8aadab998d7%2FresourceGroups%2FNew...' and the user profile 'praveenpt1580@gmail.com'.

Step 5

Select new dataset format as Delimited Text then proceed with continue.



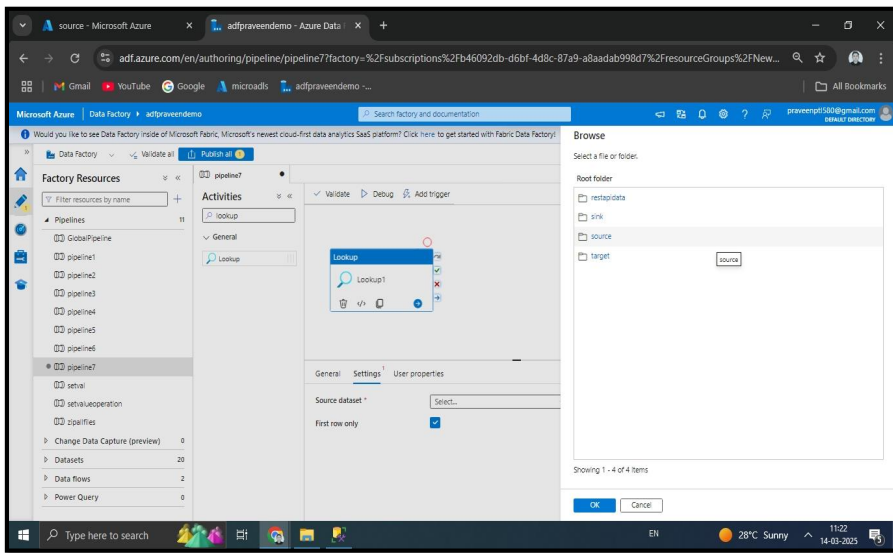
Step 6

In set properties select linked service ADLS Gen2 next click on browse button.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline7' selected. The 'Activities' pane shows a 'Lookup' activity. The main canvas displays a visual representation of the 'Lookup' activity. On the right, the 'Set properties' dialog is open. The 'Name' field is 'DelimitedText10'. The 'Linked service' dropdown is set to 'AzureDataLakeStorage1'. The 'File path' section shows 'File system' selected, with 'Directory' and 'File name' fields. The 'First row as header' checkbox is checked. The 'Import schema' section has 'From connection/store' selected. The 'Advanced' section is expanded, showing 'Source dataset' as 'Select...' and 'First row only' as checked. The 'Browse' button is visible next to the 'File name' field. At the bottom of the dialog are 'OK', 'Back', and 'Cancel' buttons.

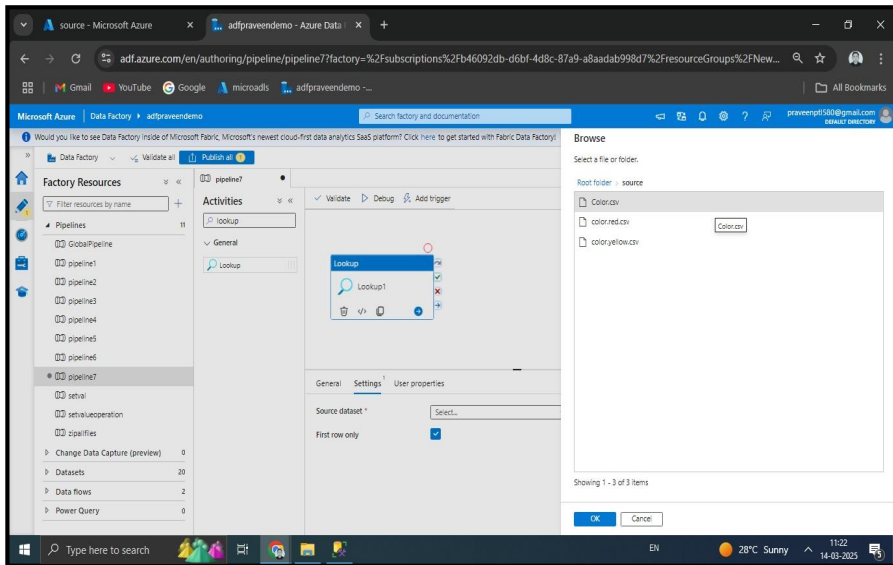
Step 7

Next go to source container to select Color.csv file.



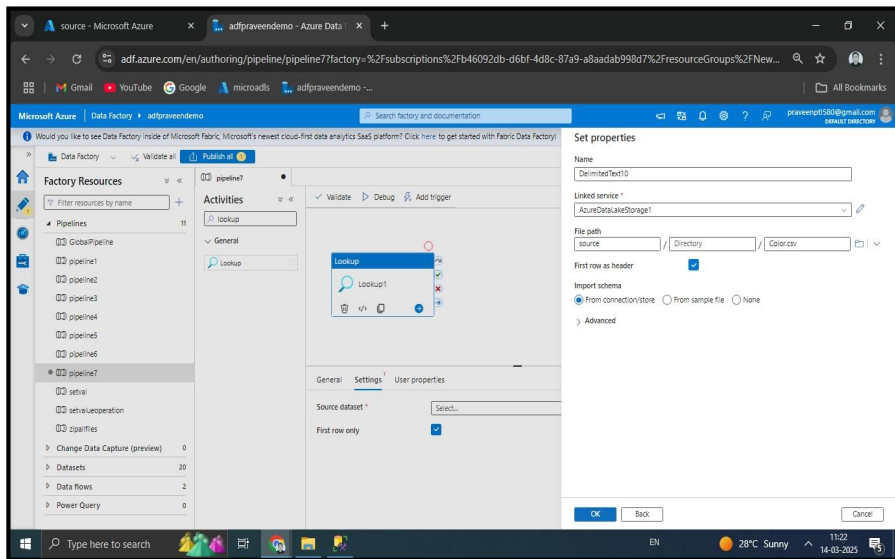
Step 8

Select Color.csv then click on ok button.



Step 9

After source container and file are browsed finally click on ok button.



Step 10

Next go to activity search bar search and select ForEach activity then drag and drop it connect lookup activity to forEach activity with green arrow.

The screenshot displays the Microsoft Azure Data Factory portal interface. The browser address bar shows the URL: adf.azure.com/en/authoring/pipeline/pipeline7?factory=%2Fsubscriptions%2Fb46092db-d6bf-4d8c-87a9-a8aadab998d7%2FresourceGroups%2FNew.... The page title is "Microsoft Azure | Data Factory | adfpravedemo". The main content area shows the "pipeline7" editor. On the left, the "Factory Resources" pane lists various resources, including "pipelines" (GlobalPipeline, pipeline1, pipeline2, pipeline3, pipeline4, pipeline5, pipeline6, pipeline7) and "Data flows" (Power Query). The "Activities" pane on the right shows the "ForEach" activity selected. The main canvas displays a "Lookup" activity connected to a "ForEach" activity with a green arrow. The "ForEach" activity is currently empty, showing "Activities: No activities". The "Properties" panel on the right shows the "General" tab with "Name: pipeline7" and "Description" field. The "Activities" panel on the left shows the "ForEach" activity selected. The bottom status bar shows the Windows taskbar with the search bar, task icons, and system tray information (EN, 28°C Sunny, 11:24, 14-03-2025).

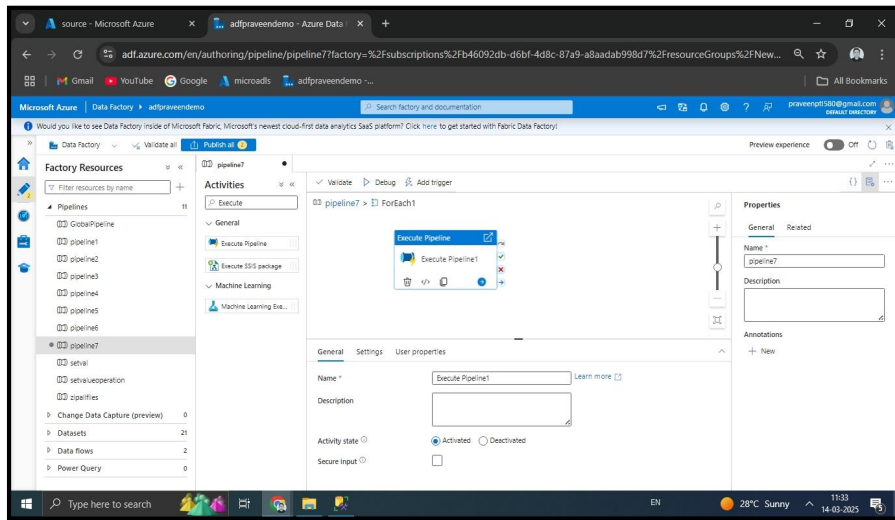
Step 11

Go to ForEach activity settings go to Items add dynamic content then go to activity outputs select 'Lookup1 value array' next click on ok button.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline7' selected. The main workspace shows a pipeline diagram with a 'Lookup' activity connected to a 'ForEach' activity. The 'ForEach' activity is highlighted, and its settings are visible in the bottom pane. The 'Settings' tab is active, showing the 'Items' property, which is currently empty. The 'Pipeline expression builder' is open on the right, showing the expression '@activity(''Lookup1'').output.value' in the 'Add dynamic content' field. Below the expression builder, the 'Activity outputs' tab is selected, showing a list of outputs: 'Lookup1', 'Lookup1 activity output', 'Lookup1 count', 'Count of the rows', and 'Lookup1 value array'. The 'Lookup1 value array' is highlighted, indicating it is the selected output for the 'Items' property.

Step 12

Go to FoEach activities click on pencil icon next go to activity search bar search and select execute pipeline activity then drag and drop it.



Step 13

Next go to execute pipeline activity click on '+ New' to create a new pipeline to invoke.

The screenshot displays the Microsoft Azure Data Factory portal interface. The left sidebar shows the 'Factory Resources' tree with 'pipelines' expanded, listing 'Global Pipeline' and 'pipeline1' through 'pipeline7'. 'pipeline7' is selected. The main canvas shows the 'Execute Pipeline' activity within a 'ForEach1' loop. The activity is configured with 'Name *' as 'pipeline7' and 'Description' as an empty field. The 'Settings' tab is active, showing 'Invoked pipeline *' as 'Select...' and 'Wait on completion' checked. The bottom status bar indicates the time as 11:33 on 14-03-2025.

Step 14

Go to pipeline parameters create a new parameter make name Color.

The screenshot displays the Microsoft Azure Data Factory portal interface. The left-hand navigation pane shows the 'Factory Resources' tree, where 'pipeline8' is selected under the 'Pipelines' folder. The main workspace is titled 'pipeline8' and contains an 'Activities' list on the left and a 'Parameters' tab on the right. The 'Parameters' tab is active, showing a table with one parameter: 'Color', which is of type 'String' and has a 'Value' field. The 'Properties' pane on the far right shows the 'General' tab for the selected parameter, with the 'Name' field set to 'pipeline8' and a 'Description' field below it. The top of the browser window shows the URL 'adf.azure.com/en/authoring/pipeline/pipeline8?factory=%2Fsubscriptions%2Fb46092db-d6bf-4d8c-87a9-a8aadab998d7%2FresourceGroups%2FNew...' and the user 'praveenpt580@gmail.com' is logged in.

Name	Type	Default value
Color	String	Value

Step 15

Go back to execute pipeline activity then go to Color add dynamic content next go to ForEachIterator select ForEach1(Current Item) write .Color after item() enclose it with opening and closing parenthesis to do string interpolation next click on ok button.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline7' selected. The 'Activities' pane shows the 'Execute' activity for 'pipeline7', which is configured to run 'pipeline8'. The 'Settings' tab is active, showing the 'Wait on completion' checkbox checked. The 'Parameters' table lists a parameter named 'Color' of type 'string'. The 'Value' column for 'Color' is currently empty, and the 'Add dynamic content' button is visible. The 'Pipeline expression builder' pane on the right is open, showing the expression '@{item().color}' entered in the text area. The 'ForEach Iterator' tab is selected, and the 'Current item' is listed as 'ForEach1'. The 'OK' button is visible at the bottom of the expression builder.

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Factory Resources

- Pipelines (12)
 - GlobaPipeline
 - pipeline1
 - pipeline2
 - pipeline3
 - pipeline4
 - pipeline5
 - pipeline6
 - pipeline7
 - pipeline8
 - setval
 - setvalueoperation
 - zipalifies
- Change Data Capture (preview) 0
- Datasets 21
- Data flows 2
- Power Query

Activities

- General
 - Execute
 - Execute Pipeline
 - Execute SSIS package
 - Machine Learning

pipeline7 > ForEach1

Execute Pipeline1 pipeline8

General Settings User properties

Invoked pipeline * pipeline8

Wait on completion ☒

Parameters

Name	Type	Value
Color	string	

Add dynamic content (Alt+S)

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

@{item().color}

Clear contents

ForEach Iterator Activity outputs Parameters System variables

Search

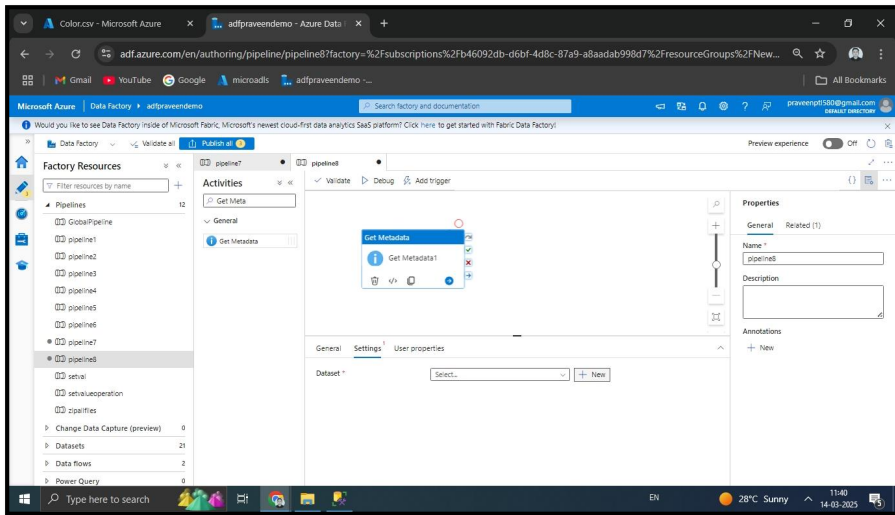
ForEach1 Current item

OK Cancel

EN 28°C Sunny 11:37 14-03-2025

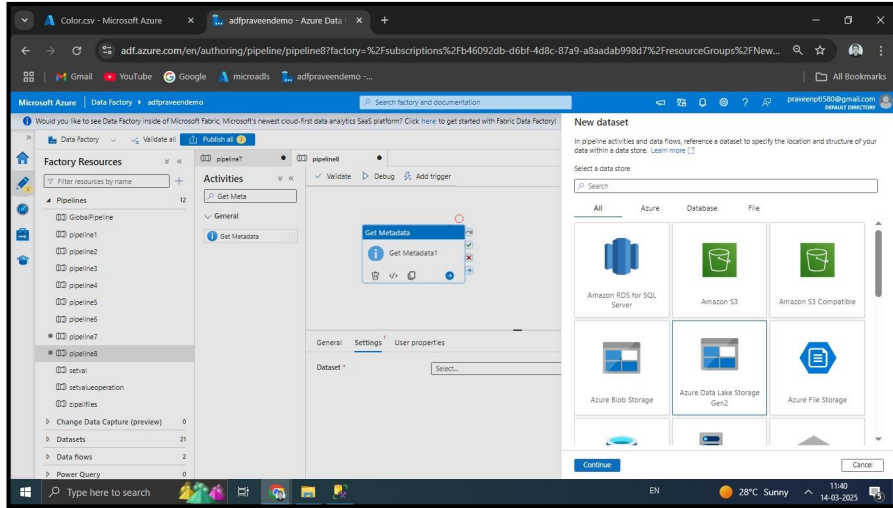
Step 16

Go to pipeline8 go to activity search bar search and select get metadata activity then drag and drop it go to get metadata settings.



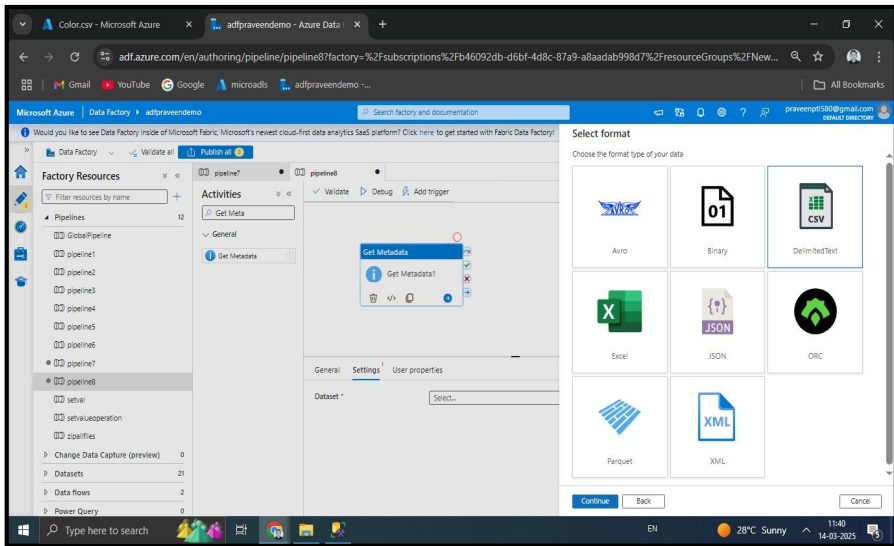
Step 17

Next click on '+ New' to create a new dataset select new dataset for ADLS Gen2 next click on continue.



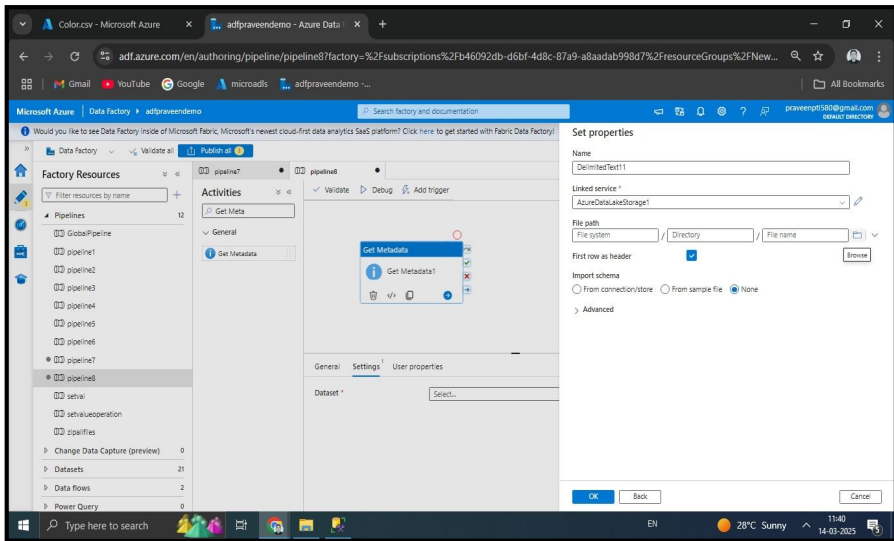
Step 18

Select new dataset format as Delimited Text then proceed with continue.



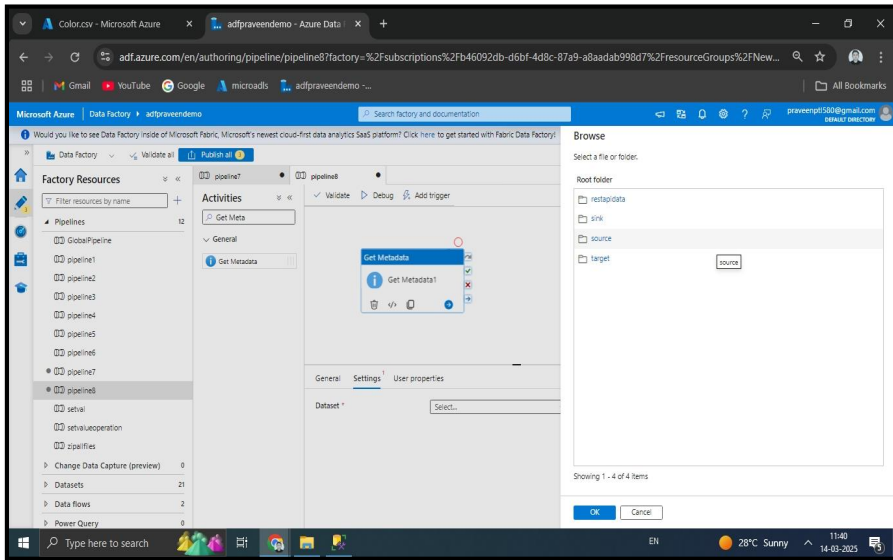
Step 19

In set properties select linked service ADLS Gen2 next click on browse button.



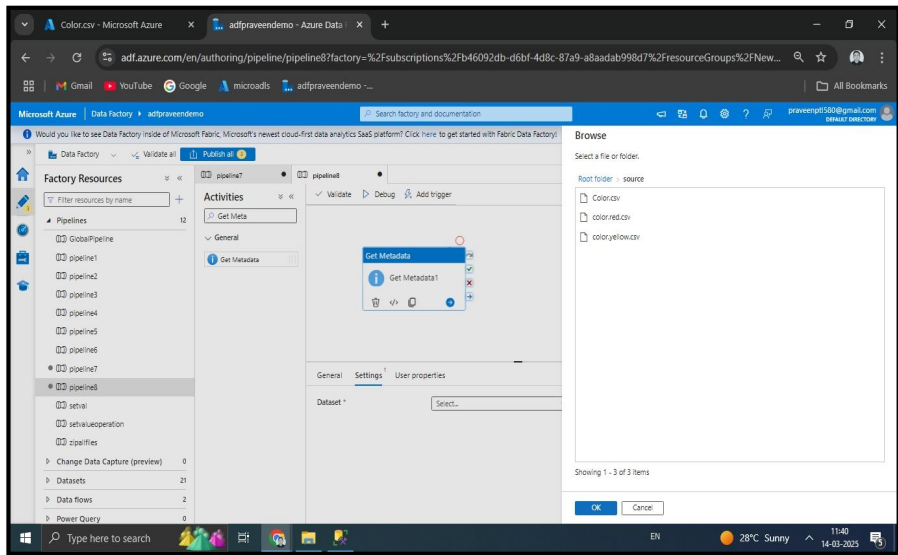
Step 20

Next go to source container.



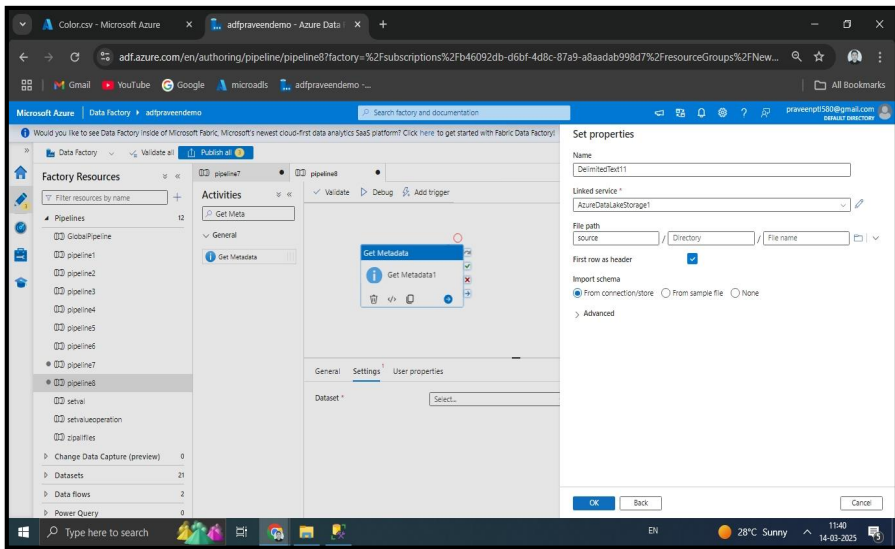
Step 21

Don't select any file then click on ok.



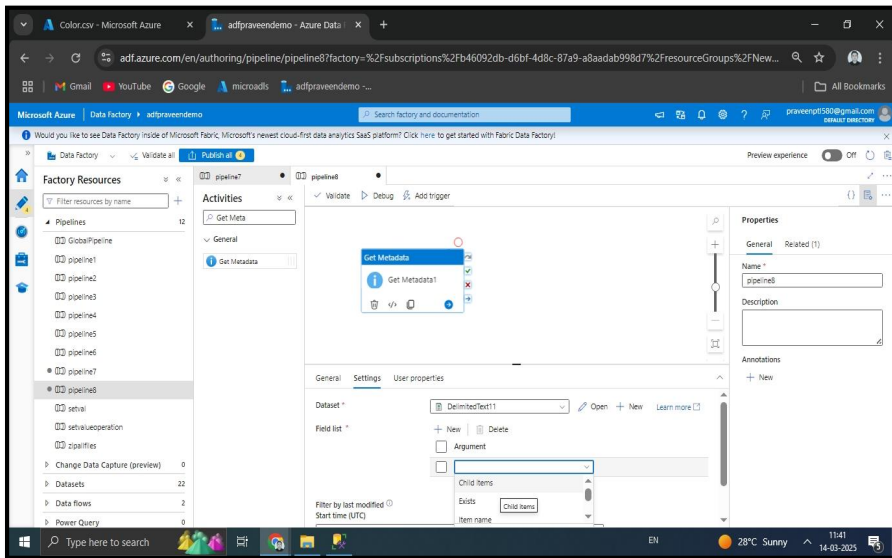
Step 22

After source container is browsed finally click on ok button.



Step 23

Next click on '+ New' to create a new field list select new field list as child items.



Step 24

Next go to activity search bar search and select **ForEach** activity then drag and drop it. Connect **get metadata** activity to **forEach** activity with green arrow.

The screenshot displays the Microsoft Azure Data Factory portal interface. The browser address bar shows the URL: adf.azure.com/en/authoring/pipeline/pipeline8?factory=%2Fsubscriptions%2Fb46092db-d6bf-4d8c-87a9-a8aadab998d7%2FresourceGroups%2FNew.... The page title is "Microsoft Azure | Data Factory | adfpravedemo".

The interface is divided into several sections:

- Factory Resources:** A sidebar on the left showing a list of pipelines (pipeline1 through pipeline8) and other resources like GlobalPipeline, setval, setvaloperation, zipallfiles, Change Data Capture (preview), Datasets, Data flows, and Power Query.
- Activities:** A central pane showing a search bar with "ForEach" entered. Below the search bar, a list of activities is displayed, including "ForEach". A "ForEach" activity is being added to the pipeline, indicated by a green arrow pointing to the "Activities" section.
- Properties:** A pane on the right showing the configuration for the selected "ForEach" activity. The "Name" field is set to "ForEach1". The "Description" field is empty. The "Activity state" is set to "Activated".

The bottom of the screen shows a Windows taskbar with a search bar and various application icons. The system tray on the right indicates the time is 11:42 on 14-03-2025, and the weather is 28°C Sunny.

Step 25

Go to ForEach activity settings click on items add dynamic content go to activity outputs select 'Get Metadata 1 childitems' then click on ok button.

The screenshot shows the Microsoft Azure Data Factory Pipeline Expression Builder interface. The left sidebar displays the 'Factory Resources' tree, with 'pipeline8' selected under the 'Pipelines' folder. The main canvas shows a pipeline diagram with a 'Get Metadata' activity (labeled 'Get Metadata1') connected to a 'ForEach' activity (labeled 'ForEach1'). The 'ForEach' activity is currently selected, and its settings are visible in the bottom pane. The 'Settings' tab is active, showing the 'Items' property, which is currently empty. The 'Add dynamic content' button is visible at the bottom of the 'Items' field. The right pane, titled 'Pipeline expression builder', shows the 'Add dynamic content below using any combination of expressions, functions and system variables.' section. The expression '@activity(''Get Metadata1'').output.childitems' is entered in the text area. Below the text area, the 'Clear contents' button is visible. The bottom pane shows the 'Activity outputs' tab, which lists the outputs of the 'Get Metadata1' activity: 'Get Metadata1 activity output', 'Get Metadata1 childitems' (List of subfolders and files in the given folder), 'Get Metadata1 exists' (Whether a file, folder, or table exists), 'Get Metadata1 itemName' (Name of the file or folder), and 'Get Metadata1 itemType'. The 'OK' and 'Cancel' buttons are at the bottom of this pane.

Step 26

Go to ForEach activities click on ForEach pencil icon.

The screenshot displays the Microsoft Azure Data Factory portal interface. The browser address bar shows the URL: `adf.azure.com/en/authoring/pipeline/pipeline8?factory=%2Fsubscriptions%2Fb46092db-d6bf-4d8c-87a9-a8aadab998d7%2FresourceGroups%2FNew...`. The page title is "Microsoft Azure | Data Factory | adfpraveendemo".

The left sidebar shows the "Factory Resources" tree with "Pipelines" expanded, listing various pipelines including "pipeline8". The main canvas displays the "pipeline8" diagram. It contains a "Get Metadata" activity followed by a "ForEach" activity. The "ForEach" activity is currently in edit mode, indicated by a pencil icon in its top right corner. The "Activities" pane on the left shows "ForEach" under the "Iteration & conditionals" category. The "Properties" pane on the right shows the "General" tab for the "ForEach" activity, with the "Name" field set to "pipeline8".

The bottom of the screen shows the Windows taskbar with the search bar and system tray icons.

Step 27

Go to activity search bar search and select ifcondition activity next drag and drop it.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline8' selected. The main workspace shows 'pipeline8' with a 'ForEach' loop containing an 'If Condition' activity. The 'If Condition' activity is currently being configured, with a search bar at the top of the workspace showing 'ifcondition'. The 'Properties' pane on the right shows the 'General' tab for the 'If Condition' activity, with the 'Name' field set to 'If Condition1'. The 'Activity state' is set to 'Activated'.

Step 28

Go to if condition activities go to expression add dynamic content write the following condition in the expression builder to ensure .csv file name substring contains Color.csv file column values next click on continue.

The screenshot shows the Microsoft Azure Data Factory Pipeline Expression Builder interface. The browser address bar displays the URL: `adf.azure.com/en/authoring/pipeline/pipeline8?factory=%2Fsubscriptions%2Fb46092db-d6bf-4d8c-87a9-a8aadab998d7%2FresourceGroups%2FNew...`. The interface includes a left sidebar with 'Factory Resources' and a list of pipelines (GlobalPipeline, pipeline1 through pipeline8, setai, setvalueoperation, zipallfiles). The main workspace shows 'pipeline8' with a 'ForEach1' activity containing an 'If Condition' activity. The 'If Condition' activity is expanded, showing a table with 'True' and 'False' outcomes, both leading to 'No activities'. The 'Expression' field is highlighted, and the 'Pipeline expression builder' dialog is open on the right. The dialog contains the following expression: `@contains(string(item().name),string(pipeline().parameters.Color))`. Below the expression field, there are tabs for 'ForEach Iterator', 'Activity outputs', 'Parameters', and 'System variables'. The 'Parameters' tab is selected, showing a search bar and a list of parameters: 'Color', 'RootFolder', and 'Global parameter'. The 'OK' and 'Cancel' buttons are at the bottom of the dialog.

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Factory Resources

- Pipelines (12)
 - GlobalPipeline
 - pipeline1
 - pipeline2
 - pipeline3
 - pipeline4
 - pipeline5
 - pipeline6
 - pipeline7
 - pipeline8
 - setai
 - setvalueoperation
 - zipallfiles
- Change Data Capture (preview) 0
- Datasets 22
- Data flows 2
- Power Query 0

Activities

- Iteration & conditionals
 - If Condition

pipeline8 > ForEach1

If Condition

True	False
No activities	No activities

General Activities (0) User properties

Expression *

This property should be parameterized.

Add dynamic content (Alt+Shift+D)

Case

True	False
No activities	No activities

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

```
@contains(string(item().name),string(pipeline().parameters.Color))
```

Clear contents

ForEach Iterator Activity outputs Parameters System variables

Search

Color
RootFolder
Global parameter

OK Cancel

Step 29

Next go to True case pencil icon.

The screenshot displays the Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline8' selected. The main workspace shows the 'pipeline8' configuration, specifically the 'ForEach1' activity. Within 'ForEach1', an 'If Condition' activity is configured. The 'If Condition' activity has a single condition named 'If Condition1'. The 'True' branch of the condition is currently empty, and a pencil icon is visible next to it, indicating it is the focus of the current step. The 'False' branch is also empty. The 'Properties' pane on the right shows the 'General' tab for the 'If Condition' activity, with the 'Name' set to 'If Condition1' and a description field. The 'Expression' field in the 'Activities (0)' tab is set to '@contains(string(item().name),string...)'. The bottom of the screen shows the Windows taskbar with the search bar, system tray, and date/time (12:00, 14-03-2025).

Step 30

Go to activity search bar search and select copy activity then drag and drop it.

The screenshot displays the Microsoft Azure Data Factory portal interface. The browser address bar shows the URL: `adf.azure.com/en/authoring/pipeline/pipeline8?factory=%2Fsubscriptions%2Fb46092db-d6bf-4d8c-87a9-a8aadab998d7%2FresourceGroups%2FNew...`. The page title is "Microsoft Azure | Data Factory | adfpravedemo".

On the left, the "Factory Resources" pane shows a list of pipelines, with "pipeline8" selected. The "Activities" pane in the center shows a search bar with "copy" entered, and a "Copy data" activity is visible in the "Move and transform" section. The main canvas shows a pipeline diagram with a "Copy data" activity being added to the "True" branch of an "If Condition1" trigger.

The "Properties" pane on the right shows the "General" tab for the selected activity, with the "Name" field set to "pipeline8". The "Source dataset" field is currently empty, with a "Select..." dropdown and a "+ New" button.

The bottom of the screen shows the Windows taskbar with the search bar, task icons, and system tray information including the date "14-03-2025" and time "12:00".

Step 31

Go to copy activity source click on '+ New' select new dataset for Azure Data Lake Gen2
Next click on continue.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline8' selected. The main workspace shows a 'Copy data' activity within 'pipeline8'. The 'Source' tab is active, and the 'Source dataset' dropdown is open, showing 'Azure Data Lake Gen2' as the selected option. The right-hand pane, titled 'New dataset', provides instructions on how to specify a dataset and offers a search bar to select a data store. Below the search bar, a list of data stores is shown, with 'Azure Data Lake Storage Gen2' highlighted. At the bottom of the right pane, there are 'Continue' and 'Cancel' buttons. The top of the browser window shows the URL 'adf.azure.com/en/authoring/pipeline/pipeline8?factory=%2Fsubscriptions%2Fb46092db-d6bf-4d8c-87a9-a8aadab998d7%2FresourceGroups%2FNew...'.

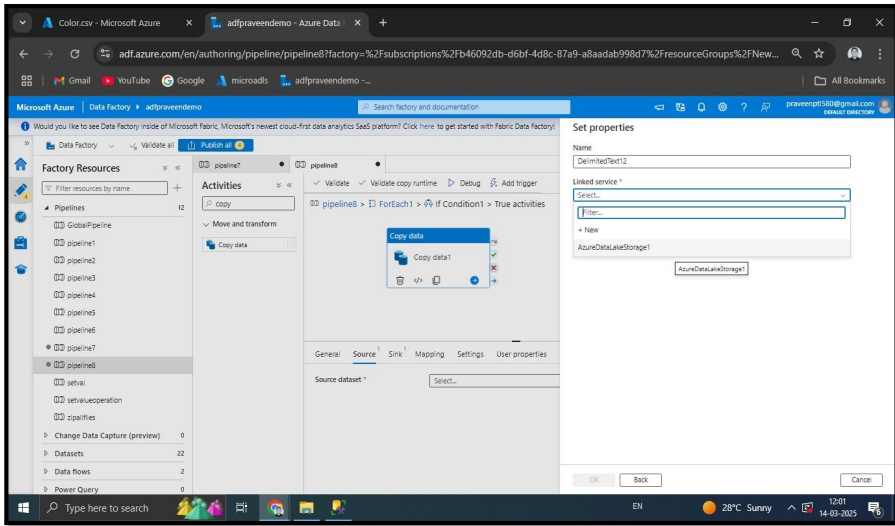
Step 32

Select new dataset format as Delimited Text then proceed with continue.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline8' selected. The main workspace shows the 'Activities' pane for 'pipeline8', containing a 'Copy data' activity. The 'Source' tab is active, showing a 'Source dataset' dropdown menu. On the right, the 'Select format' dialog is open, prompting the user to 'Choose the format type of your data'. The dialog displays a grid of format options: Avro, Binary, DelimitedText (highlighted), Excel, JSON, ORC, Parquet, and XML. At the bottom of the dialog are 'Continue', 'Back', and 'Cancel' buttons. The bottom of the screen shows the Windows taskbar with the search bar, system tray, and date/time (12:01, 14-03-2025).

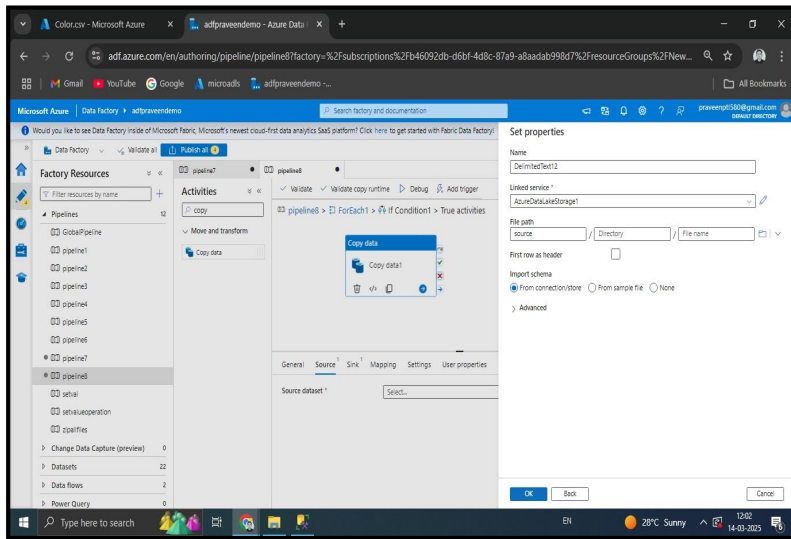
Step 33

In set properties select linked service ADLS Gen2 next click on browse button to browse source container.



Step 34

Go to source container don't select any file click on ok after source container is browsed finally click on ok.



Step 35

After dataset is created click on open pencil icon.

The screenshot displays the Microsoft Azure Data Factory portal interface. The browser address bar shows the URL: `adf.azure.com/en/authoring/pipeline/pipeline8?factory=%2Fsubscriptions%2Fb46092db-d6bf-4d8c-87a9-a8aadab998d7%2FresourceGroups%2FNew...`. The page title is "Microsoft Azure | Data Factory | adfpraveendemo".

The left sidebar shows the "Factory Resources" tree with the following structure:

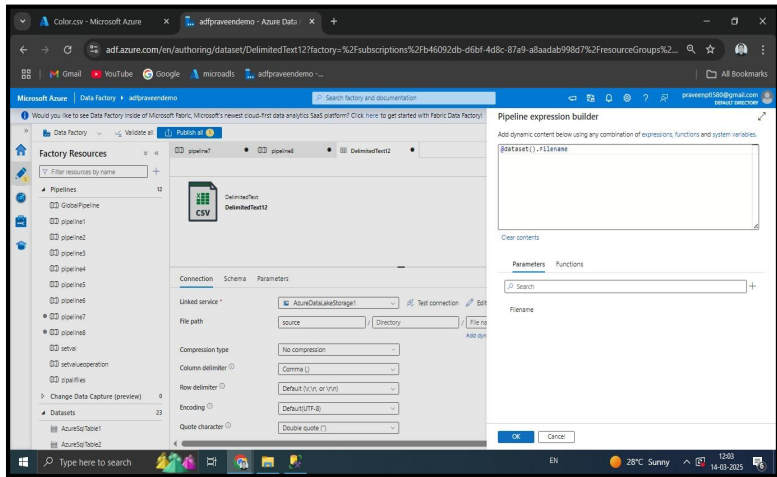
- Pipelines (12)
 - GlobalPipeline
 - pipeline1
 - pipeline2
 - pipeline3
 - pipeline4
 - pipeline5
 - pipeline6
 - pipeline7
 - pipeline8** (selected)
 - setval
 - setvalueoperation
 - zipallfiles
- Change Data Capture (preview) (0)
- Datasets (23)
- Data flows (2)
- Power Query (0)

The main workspace shows the "Activities" pane for "pipeline8". The "Copy data" activity is selected, and its configuration is shown in the "Properties" pane. The "Source" tab is active, showing the "Source dataset" as "DelimitedText12". The "File path type" is set to "File path in dataset". The "Filter by last modified" section shows "Start time (UTC)" and "End time (UTC)" input fields.

The right sidebar shows the "Properties" pane for the selected activity, with the "Name" field set to "pipeline8".

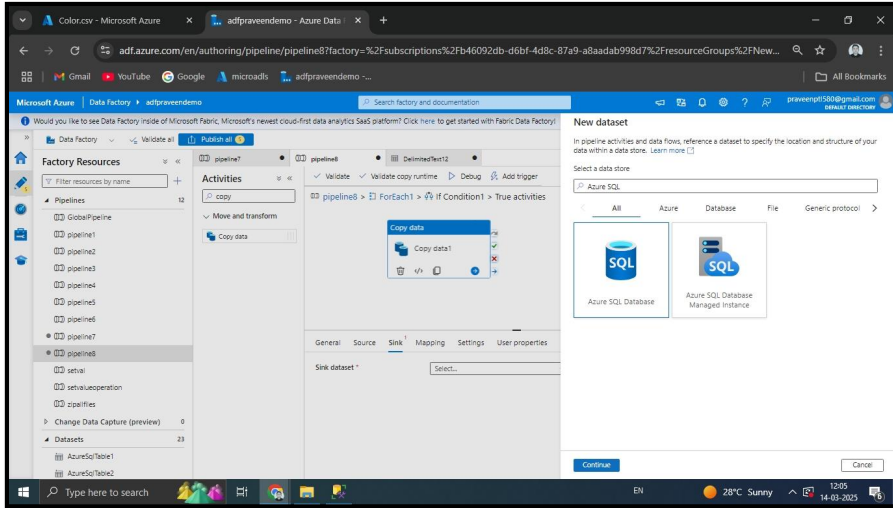
Step 36

After dataset is created click on open pencil icon. Go to parameters create a new parameter Filename next go to connections click on Filename filepath add dynamic content go to parameters select Filename then click on ok.



Step 37

Now go to copy activity sink click on '+ New' to create a new sink dataset next select new sink dataset for Azure SQL Database next click on continue.



Step 38

In set properties select linked service for Azure SQL Database next click on ok.

The screenshot displays the Microsoft Azure Data Factory web interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline8' selected. The main workspace shows the 'Copy data' activity within 'pipeline8'. A 'Set properties' dialog box is open on the right side of the screen. In this dialog, the 'Name' field is 'AzureSqlTables'. The 'Linked service' dropdown is set to 'NewAzureSqlDatabase'. The 'Table name' dropdown is set to 'Select...'. The 'Import schema' section has 'From connection/store' selected. The 'Sink dataset' dropdown is also set to 'Select...'. At the bottom of the dialog, there are 'OK', 'Back', and 'Cancel' buttons. The background interface shows the 'Activities' pane with 'Copy data' selected, and the 'Sink' tab is active in the bottom pane.

Step 39

Open sink dataset go to parameters create a new parameter tablename.

The screenshot shows the Microsoft Fabric Data Factory portal interface. On the left, the 'Factory Resources' pane lists various pipelines and datasets. The main area displays the configuration for a dataset named 'AzureSqlTable5', which is connected to an 'Azure SQL Database'. The 'Parameters' tab is active, showing a table with columns for Name, Type, and Default value. A new parameter named 'tablename' is being added with a 'String' type and a 'Value' default.

Name	Type	Default value
tablename	String	Value

Step 40

Go back to connections write table schema as dbo go to tablename add dynamic content select parameter tablename next click on ok.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline7' selected. The main workspace shows the configuration for 'pipeline7', specifically the 'Connection' tab for the 'AzureSqlTable5' dataset. The 'Linked service' is set to 'NewAzureSqlDatabase'. The 'Table' is set to 'dbo', and the 'Enter manually' checkbox is checked. The 'Parameters' tab is active, showing a search bar and a list of parameters, including 'tablename'. The 'Pipeline expression builder' pane on the right shows the dynamic content '@dataset().tablename'.

Step 41

Now go to sink go to tablename add dynamic content write `@{item.name()}` then click on ok.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline8' selected. The main workspace shows the 'Copy data' activity within 'pipeline8'. The 'Sink' tab is active, showing the 'Sink dataset' as 'AzureSqlTable5'. Under 'Dataset properties', the 'Name' is 'tablename', and the 'Add dynamic' button is visible. The 'Write behavior' is set to 'Insert'. On the right, the 'Pipeline expression builder' is open, showing the dynamic content '@{item().name()}' in the text area. Below the text area, the 'Clear contents' button is visible. The 'Activity outputs' tab is selected, showing a list of metadata outputs including 'Get Metadata1', 'Get Metadata1 childrens', 'Get Metadata1 exists', 'Get Metadata1 itemName', and 'Get Metadata1 itemType'. The 'OK' button is highlighted at the bottom of the expression builder.

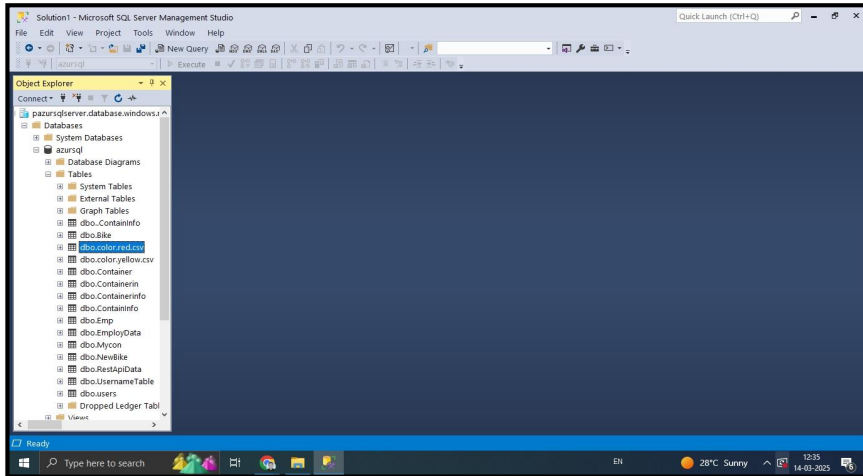
Step 42

Now execute pipeline using debug mode once pipeline executes successfully it shows execution status as succeeded.

The screenshot displays the Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline7' selected. The main workspace is titled 'pipeline7' and shows the 'Debug' mode for a specific activity named 'ForEach1'. A tooltip indicates that the trigger test runs the current pipeline without publishing changes. The 'Activities' pane on the left lists 'Execute', 'Execute Pipeline', 'Execute SSIS package', and 'Machine Learning'. The 'Properties' pane on the right shows the 'General' tab for 'ForEach1', with fields for 'Name' (set to 'ForEach1') and 'Description'. The 'Activity state' is set to 'Activated'. The bottom of the screen shows the Windows taskbar with the search bar, system clock (12:15, 14-03-2025), and weather (28°C Sunny).

Step 43

Now go back to microsoft sql server studio login it with authentication credentials like server name , authentication etc. Server opened click on refresh go to database under dataset you get to see two tables `dbo.color.red.csv` and `dbo.color.yellow` whose substring match with Color column value red and yellow.





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